surveillance, agriculture, infrastructure inspection, and environmental monitoring.

In addition to perception and navigation, AI has improved mission planning and task execution. Algorithms can now optimize flight routes to conserve energy, avoid hazards, and complete objectives efficiently. Multi-drone coordination systems enable fleets of UAVs to divide complex tasks, share information, and cooperate as functional units, whether mapping large geographic areas or conducting search-and-rescue operations. As AI continues to advance, drones are expected to gain greater operational intelligence, adaptability, and collaborative capability, expanding their role in both civilian and strategic mission environments.

9.2.3 The role of machine learning and neural networks in drone optimization

Machine learning and neural networks play a central role in enhancing the performance, adaptability, and efficiency of modern drones. Unlike traditional programed systems that follow fixed instructions, machine learning enables drones to learn from experience, improve with continued use, and respond intelligently to changing environments. By analyzing large datasets such as flight histories, environmental patterns, sensor recordings, and visual imagery drones develop predictive capabilities that allow them to anticipate obstacles, adjust flight behavior, and refine operational strategies over time.

Neural networks, particularly convolutional neural networks (CNNs) and recurrent neural networks (RNNs), support advanced processing tasks such as real-time object detection, image classification, motion prediction, and route optimization. CNNs help drones recognize shapes, detect anomalies, and interpret visual scenes with high accuracy, which is crucial for activities such as surveillance, agricultural crop analysis, and infrastructure